HOSPITAL EMERGENCY QUEUE MANAGEMENT SYSTEM PROJECT REPORT

Abstract

This project develops a hospital emergency queue management system using a priority queue based on Emergency Severity Index (ESI) levels. Patients are prioritized by severity and arrival time.This project develops a hospital emergency queue management system using a priority queue based on Emergency Severity Index (ESI) levels. Patients are prioritized by severity and arrival time.This project develops a hospital emergency queue management system using a priority queue based on Emergency Severity Index (ESI) levels. Patients are prioritized by severity and arrival time.

Introduction

Emergency departments require efficient patient prioritization. The system minimizes waiting time for critical patients and improves resource utilization.Emergency departments require efficient patient prioritization. The system minimizes waiting time for critical patients and improves resource utilization.Emergency departments require efficient patient prioritization. The system minimizes waiting time for critical patients and improves resource utilization.

Problem Statement

Traditional first-come-first-served queues may delay treatment of critical patients.Traditional first-come-first-served queues may delay treatment of critical patients.Traditional first-come-first-served queues may delay treatment of critical patients.

Objectives

Implement severity-based triage, real-time dashboards, doctor allocation, notifications, and scalable deployment.Implement severity-based triage, real-time dashboards, doctor allocation, notifications, and scalable deployment.Implement severity-based triage, real-time dashboards, doctor allocation, notifications, and scalable deployment.

Literature Survey

Reviews queue management systems, triage methodologies, and priority scheduling algorithms.Reviews queue management systems, triage methodologies, and priority scheduling algorithms.Reviews queue management systems, triage methodologies, and priority scheduling algorithms.

Requirements Analysis

Functional requirements include patient registration, queue management, doctor assignment, and monitoring. Non-functional requirements include scalability, reliability, and security.Functional requirements include patient registration, queue management, doctor assignment, and monitoring. Non-functional requirements include scalability, reliability, and security.Functional requirements include patient registration, queue management, doctor assignment, and monitoring. Non-functional requirements include scalability, reliability, and security.

System Architecture

API Gateway, Backend Service, Priority Queue Engine, PostgreSQL Database, React Frontend, Monitoring Layer.API Gateway, Backend Service, Priority Queue Engine, PostgreSQL Database, React Frontend, Monitoring Layer.API Gateway, Backend Service, Priority Queue Engine, PostgreSQL Database, React Frontend, Monitoring Layer.

Database Design

Patients, Doctors, and Triage Levels tables with relationships and constraints.Patients, Doctors, and Triage Levels tables with relationships and constraints.Patients, Doctors, and Triage Levels tables with relationships and constraints.

Algorithm Design

Binary Min-Heap priority queue with severity and arrival-time ordering.Binary Min-Heap priority queue with severity and arrival-time ordering.Binary Min-Heap priority queue with severity and arrival-time ordering.

Technology Stack

Node.js, Express, React, Tailwind CSS, PostgreSQL, Docker, Kubernetes.Node.js, Express, React, Tailwind CSS, PostgreSQL, Docker, Kubernetes.Node.js, Express, React, Tailwind CSS, PostgreSQL, Docker, Kubernetes.

Testing Strategy

Unit testing, API testing, integration testing, and performance testing.Unit testing, API testing, integration testing, and performance testing.Unit testing, API testing, integration testing, and performance testing.

Expected Results

Reduced patient waiting time and improved emergency response.Reduced patient waiting time and improved emergency response.Reduced patient waiting time and improved emergency response.

Future Enhancements

AI-assisted triage, predictive analytics, and mobile applications.AI-assisted triage, predictive analytics, and mobile applications.AI-assisted triage, predictive analytics, and mobile applications.

Conclusion

The system provides efficient and scalable emergency queue management.The system provides efficient and scalable emergency queue management.The system provides efficient and scalable emergency queue management.